

# Drakens Energy 360

An end-to-end synthetic data engineering platform for a downstream energy business

Report generated 2026-09-07 · Warehouse: drakens360\_test\_full.duckdb

**Independent synthetic portfolio project.** Drakens Energy is a fictional company invented for this project. It does not exist. Every site, customer, employee, supplier, asset, coordinate, price, volume, margin and incident in this report is randomly generated. Site coordinates are South African town centroids plus random jitter and do not represent any real service station.

## Summary

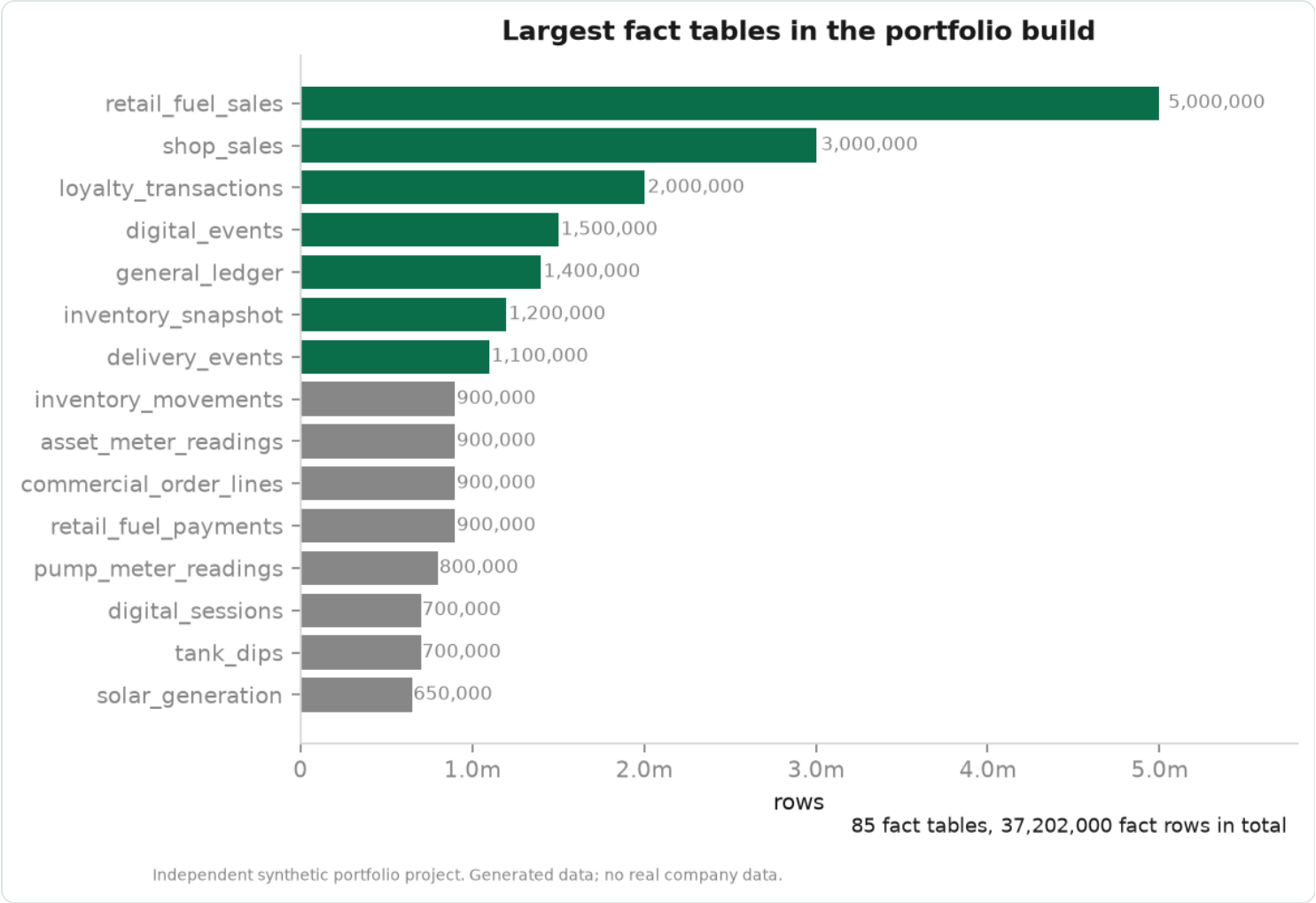
|             |            |                  |           |
|-------------|------------|------------------|-----------|
| FACT TABLES | FACT ROWS  | DEFECTS INJECTED | PROVINCES |
| 85          | 37,202,000 | 62,924,526       | 9         |

The platform covers retail forecourt, commercial B2B, supply and distribution, LPG, lubricants, aviation, marine, loyalty, digital, EV and solar, asset maintenance, HSSEQ and finance — modelled on South Africa with all nine provinces and roughly 1,050 synthetic service stations. It is built to demonstrate the parts of data engineering that are hard, not the parts that demo well: a deliberately dirty landing zone, a cleansing layer whose output is measured rather than asserted, and models reported against baselines they have to beat.

# Build

| Table                        | Rows      |
|------------------------------|-----------|
| fact_retail_fuel_sales       | 5,000,000 |
| fact_shop_sales              | 3,000,000 |
| fact_loyalty_transactions    | 2,000,000 |
| fact_digital_events          | 1,500,000 |
| fact_general_ledger          | 1,400,000 |
| fact_inventory_snapshot      | 1,200,000 |
| fact_delivery_events         | 1,100,000 |
| fact_inventory_movements     | 900,000   |
| fact_asset_meter_readings    | 900,000   |
| fact_commercial_order_lines  | 900,000   |
| fact_retail_fuel_payments    | 900,000   |
| fact_pump_meter_readings     | 800,000   |
| fact_digital_sessions        | 700,000   |
| fact_tank_dips               | 700,000   |
| fact_solar_generation        | 650,000   |
| fact_customer_invoices       | 620,000   |
| fact_site_energy_consumption | 600,000   |
| fact_customer_receipts       | 560,000   |
| fact_commercial_orders       | 550,000   |
| fact_mobile_payments         | 520,000   |
| fact_site_daily_operations   | 500,000   |
| fact_loyalty_points_balance  | 480,000   |

| Table                     | Rows    |
|---------------------------|---------|
| fact_finance_revenue_cost | 420,000 |
| fact_forecourt_shift      | 420,000 |
| fact_lpg_sales            | 400,000 |

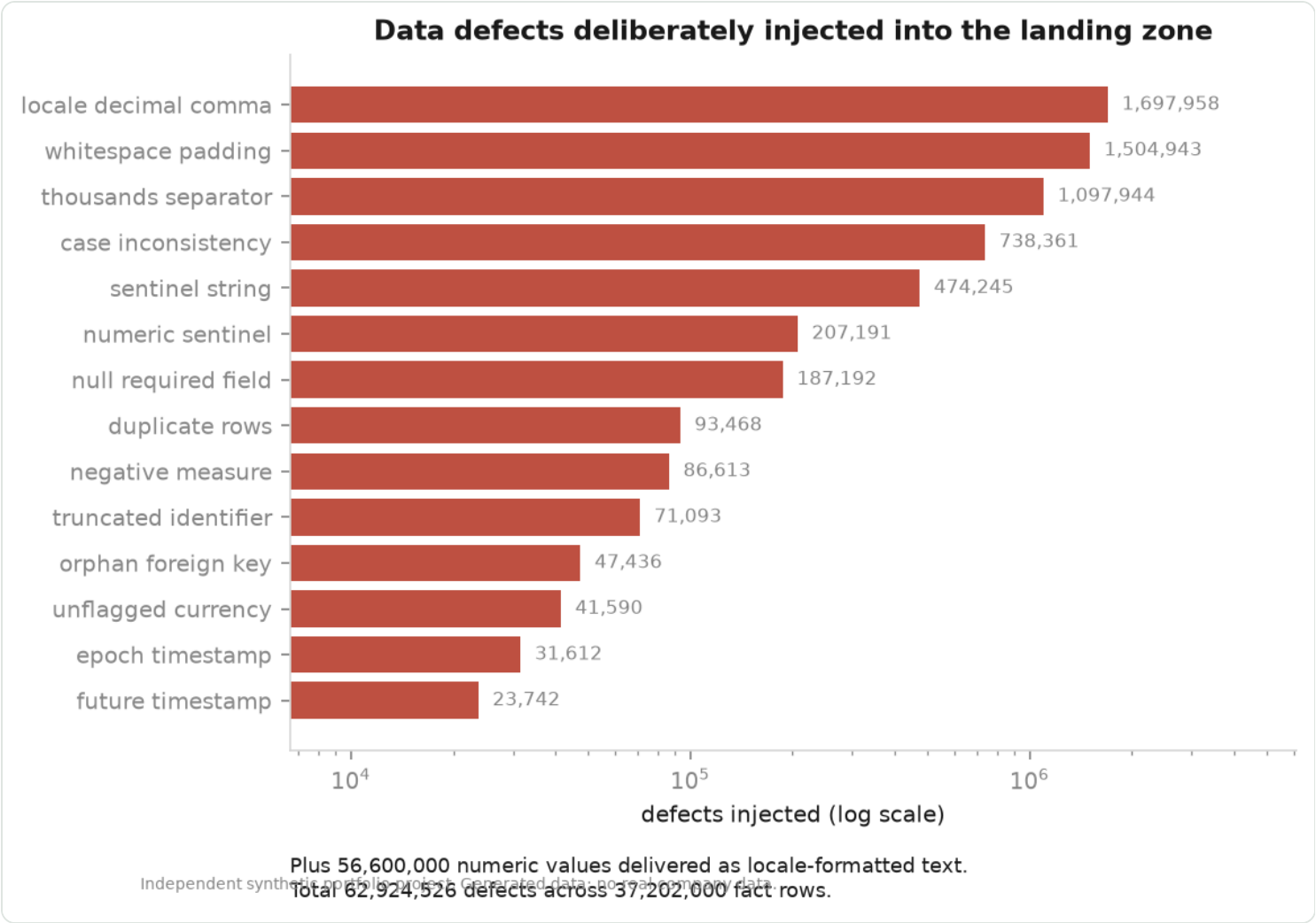


Largest fact tables in the portfolio build.

## Data quality

The landing zone is dirty on purpose. A portfolio project built on perfect data demonstrates nothing, because the difficult part of the job is deciding what to do with the row where litres reads "1 234,5", the site code has a trailing space, and the same transaction arrived twice because the point-of-sale replayed its batch. Every defect is counted as it is injected, which turns cleansing from an assertion into a measurement.

| Defect type               | Defects    |
|---------------------------|------------|
| numeric delivered as text | 56,600,000 |
| locale decimal comma      | 1,697,958  |
| whitespace padding        | 1,504,943  |
| thousands separator       | 1,097,944  |
| case inconsistency        | 738,361    |
| sentinel string           | 474,245    |
| numeric sentinel          | 207,191    |
| null required field       | 187,192    |
| duplicate rows            | 93,468     |
| negative measure          | 86,613     |
| truncated identifier      | 71,093     |
| orphan foreign key        | 47,436     |
| unflagged currency        | 41,590     |
| epoch timestamp           | 31,612     |
| future timestamp          | 23,742     |
| derived value mismatch    | 12,256     |
| unit of measure error     | 7,233      |
| encoding mojibake         | 1,649      |

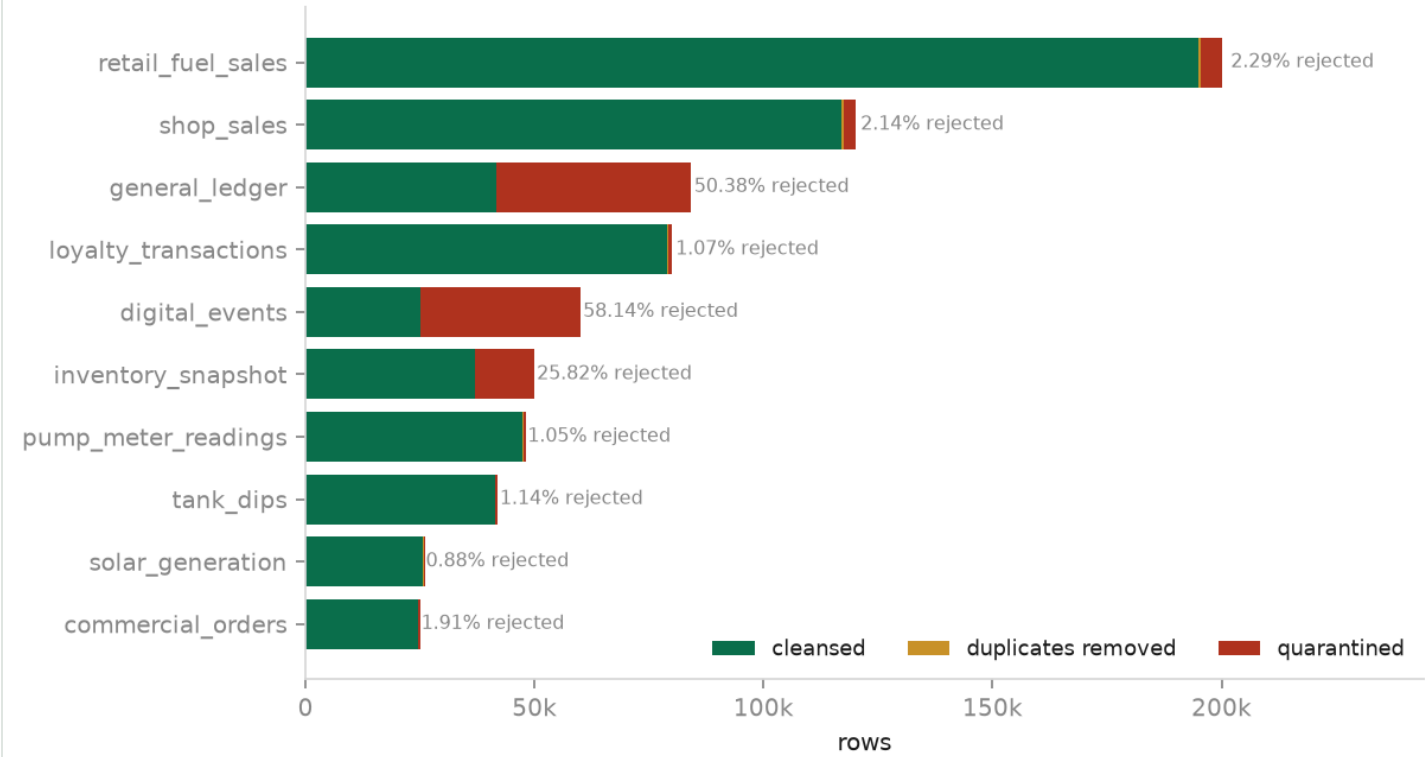


Defects deliberately injected into the landing zone.

### Cleansing outcome

| Table                        | Raw rows | Cleansed | Quarantined | Duplicates removed |
|------------------------------|----------|----------|-------------|--------------------|
| fact_retail_fuel_sales       | 200,000  | 194,934  | 4,572       | 494                |
| fact_shop_sales              | 120,000  | 117,120  | 2,572       | 308                |
| fact_general_ledger          | 84,000   | 41,579   | 42,318      | 103                |
| fact_loyalty_transactions    | 80,000   | 78,944   | 857         | 199                |
| fact_digital_events          | 60,000   | 25,054   | 34,886      | 60                 |
| fact_inventory_snapshot      | 50,000   | 37,003   | 12,912      | 85                 |
| fact_pump_meter_readings     | 48,000   | 47,384   | 502         | 114                |
| fact_tank_dips               | 42,000   | 41,416   | 478         | 106                |
| fact_solar_generation        | 26,000   | 25,704   | 228         | 68                 |
| fact_commercial_orders       | 25,000   | 24,467   | 478         | 55                 |
| fact_deliveries              | 15,000   | 14,822   | 147         | 31                 |
| fact_ev_charging_sessions    | 6,000    | 5,896    | 92          | 12                 |
| fact_maintenance_work_orders | 4,000    | 3,958    | 38          | 4                  |
| fact_hsseq_incidents         | 800      | 795      | 4           | 1                  |

Every landed row is accounted for: cleansed, deduplicated or quarantined



Drawn from drakens360\_test\_full.duckdb (194,934 retail transactions in gold). Independent synthetic portfolio project. Generated data; no real company data.

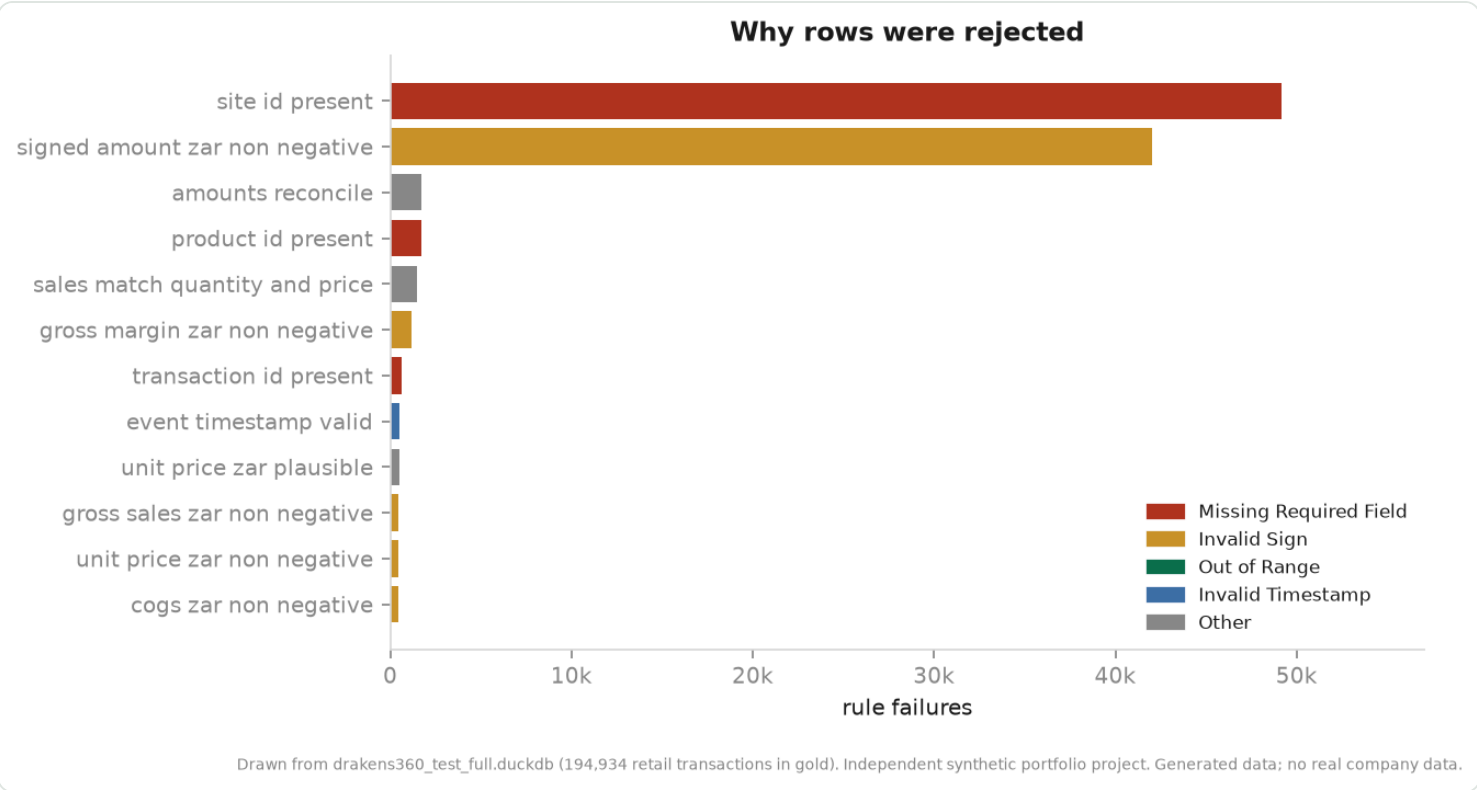
Every landed row is accounted for as cleansed, deduplicated or quarantined.

Why rows were rejected

| Rule that rejected the row     | Category               | Likely owner     | Rows   |
|--------------------------------|------------------------|------------------|--------|
| site_id_present                | Missing Required Field | Source System    | 49,166 |
| signed_amount_zar_non_negative | Invalid Sign           | Business Process | 42,046 |
| amounts_reconcile              | Other                  | Data Platform    | 1,734  |
| product_id_present             | Missing Required Field | Source System    | 1,731  |
| sales_match_quantity_and_price | Other                  | Data Platform    | 1,452  |
| gross_margin_zar_non_negative  | Invalid Sign           | Business Process | 1,166  |
| transaction_id_present         | Missing Required Field | Source System    | 630.00 |
| event_timestamp_valid          | Invalid Timestamp      | Source System    | 506.00 |
| unit_price_zar_plausible       | Other                  | Data Platform    | 497.00 |
| gross_sales_zar_non_negative   | Invalid Sign           | Business Process | 473.00 |
| unit_price_zar_non_negative    | Invalid Sign           | Business Process | 460.00 |
| cogs_zar_non_negative          | Invalid Sign           | Business Process | 458.00 |
| shop_line_id_present           | Missing Required Field | Source System    | 391.00 |
| journal_line_id_present        | Missing Required Field | Source System    | 311.00 |
| loyalty_transaction_id_present | Missing Required Field | Source System    | 284.00 |
| litres_non_negative            | Invalid Sign           | Business Process | 253.00 |
| digital_event_id_present       | Missing Required Field | Source System    | 197.00 |
| reading_id_present             | Missing Required Field | Source System    | 176.00 |
| litres_within_range            | Out of Range           | Data Entry       | 161.00 |
| snapshot_id_present            | Missing Required Field | Source System    | 160.00 |
| tank_dip_id_present            | Missing Required Field | Source System    | 125.00 |
| points_value_zar_non_negative  | Invalid Sign           | Business Process | 117.00 |
| debit_amount_zar_non_negative  | Invalid Sign           | Business Process | 93.00  |
| solar_reading_id_present       | Missing Required Field | Source System    | 91.00  |



| Rule that rejected the row        | Category               | Likely owner     | Rows  |
|-----------------------------------|------------------------|------------------|-------|
| order_id_present                  | Missing Required Field | Source System    | 91.00 |
| qualifying_spend_zar_non_negative | Invalid Sign           | Business Process | 90.00 |
| customer_id_present               | Missing Required Field | Source System    | 88.00 |
| volume_litres_non_negative        | Invalid Sign           | Business Process | 88.00 |
| credit_amount_zar_non_negative    | Invalid Sign           | Business Process | 88.00 |
| volume_litres_within_range        | Out of Range           | Data Entry       | 59.00 |
| delivery_id_present               | Missing Required Field | Source System    | 56.00 |
| revenue_zar_non_negative          | Invalid Sign           | Business Process | 43.00 |
| energy_kwh_non_negative           | Invalid Sign           | Business Process | 39.00 |
| ev_session_id_present             | Missing Required Field | Source System    | 33.00 |
| energy_kwh_within_range           | Out of Range           | Data Entry       | 32.00 |
| net_price_zar_non_negative        | Invalid Sign           | Business Process | 32.00 |
| list_price_zar_non_negative       | Invalid Sign           | Business Process | 27.00 |
| delivered_litres_non_negative     | Invalid Sign           | Business Process | 19.00 |
| delivered_litres_within_range     | Out of Range           | Data Entry       | 13.00 |
| grid_cost_zar_non_negative        | Invalid Sign           | Business Process | 8.00  |
| work_order_id_present             | Missing Required Field | Source System    | 8.00  |
| total_cost_zar_non_negative       | Invalid Sign           | Business Process | 7.00  |
| labour_cost_zar_non_negative      | Invalid Sign           | Business Process | 7.00  |
| parts_cost_zar_non_negative       | Invalid Sign           | Business Process | 3.00  |
| incident_id_present               | Missing Required Field | Source System    | 2.00  |
| estimated_cost_zar_non_negative   | Invalid Sign           | Business Process | 1.00  |



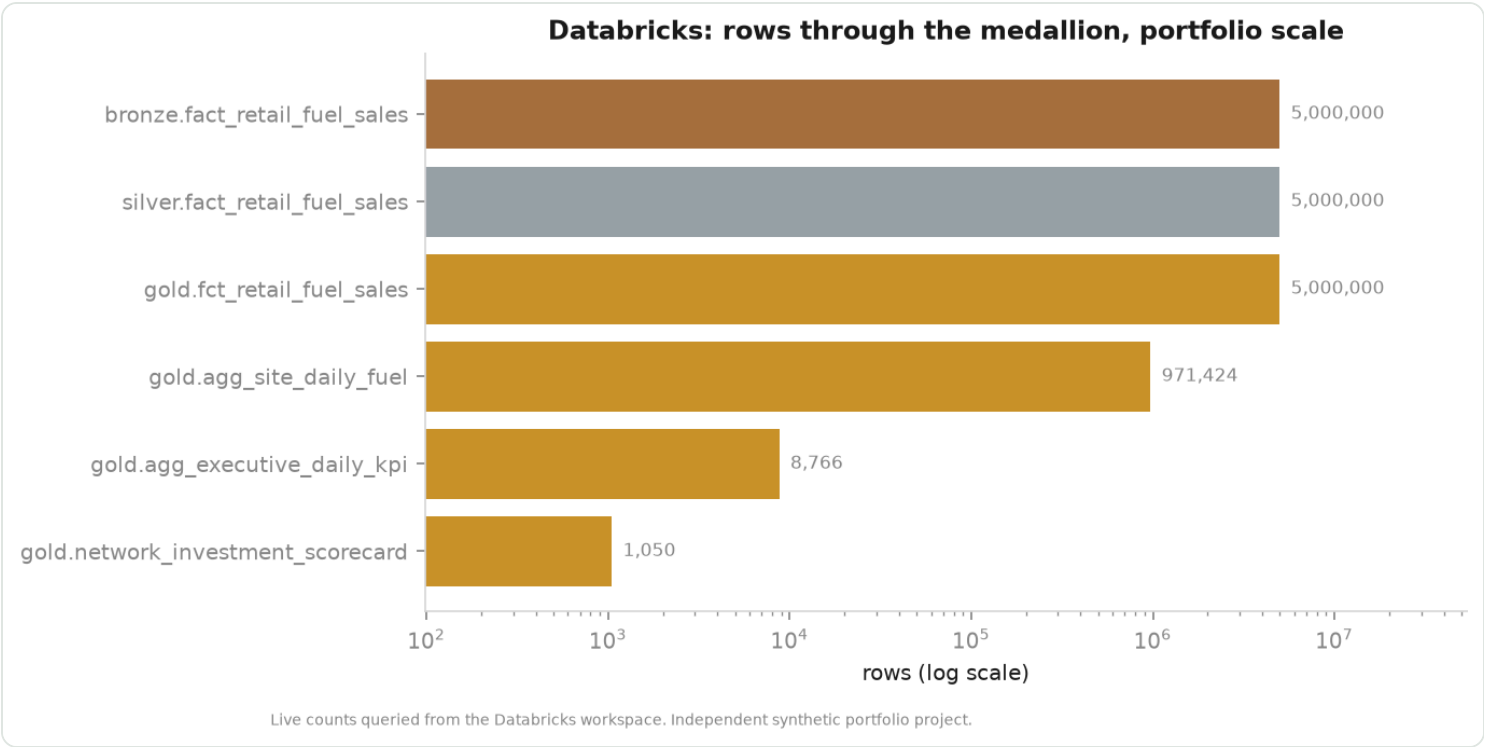
Rejection reasons. The operational response differs by rule.

## Warehouse

| Control                           | What it checks                                                 | Owner      | Variance % | Result |
|-----------------------------------|----------------------------------------------------------------|------------|------------|--------|
| DELIVERIES_HAVE_KNOWN_DESTINATION | Every delivery resolves to a site in the conformed dimension   | Supply     | 2.98       | FAIL   |
| GL_DEBITS_EQUAL_CREDITS           | Posted journal debits equal credits across the ledger          | Finance    | 99.98      | FAIL   |
| ORDERS_HAVE_KNOWN_CUSTOMER        | Every commercial order resolves to a customer in the dimension | Commercial | 2.59       | FAIL   |
| RETAIL_MARGIN_RECONCILES          | Retail fuel gross margin equals revenue less cost of sales     | Finance    | 0.00       | PASS   |
| RETAIL_VOLUME_TIES_TO_DAILY_AGG   | Transaction-level litres tie to the site-day aggregate         | Supply     | 0.00       | PASS   |

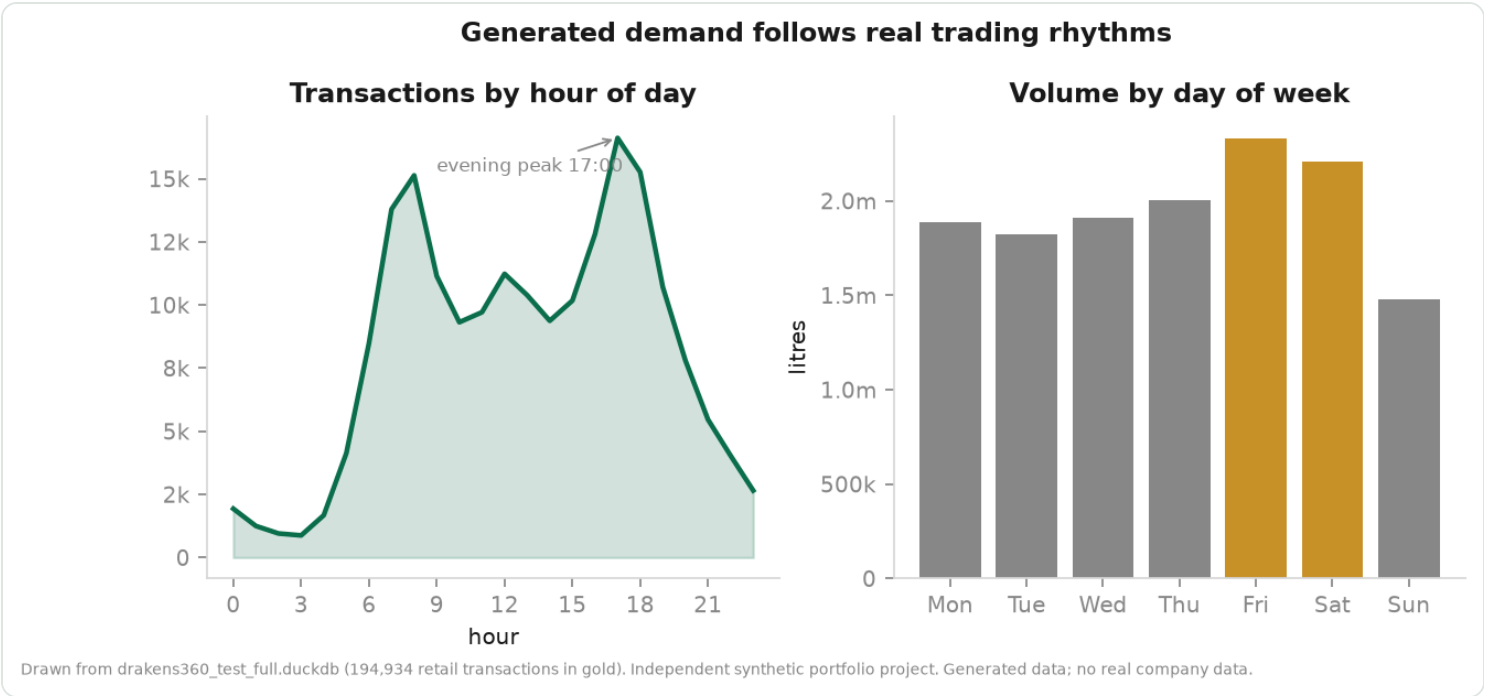
A reconciliation control evaluates a published number as data rather than only as a test, so a control that fails names the figure that is wrong and by how much. Controls that fail are shown as failures; a report that quietly

omits them would defeat the purpose of having them.



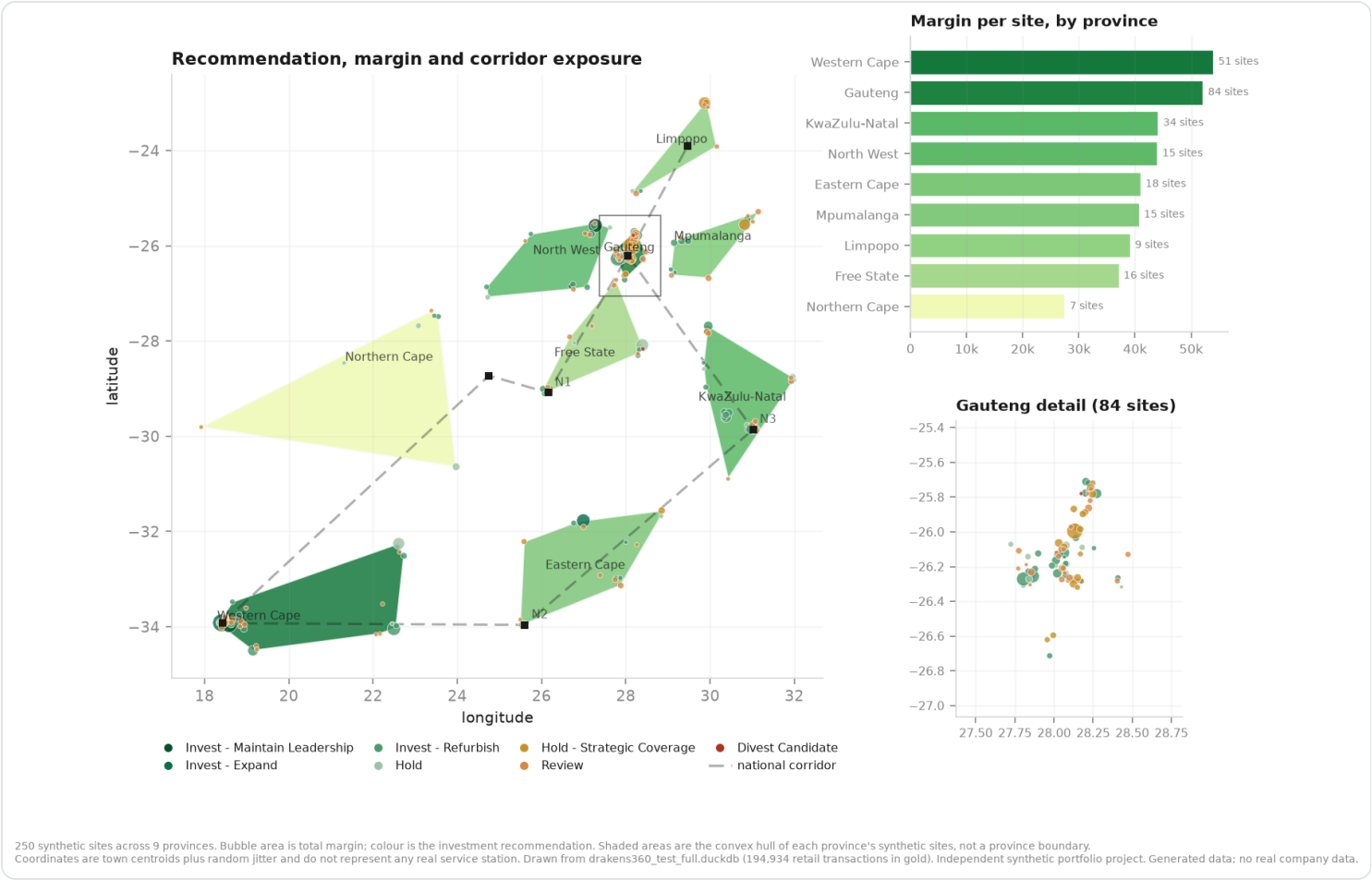
Live row counts through bronze, silver and gold on Databricks at portfolio scale.

## Demand signal



Generated demand follows real trading rhythms: commute peaks at 08:00 and 17:00, and a Friday-heavy week. Without this the forecasting model would have nothing to learn.

# Network and investment

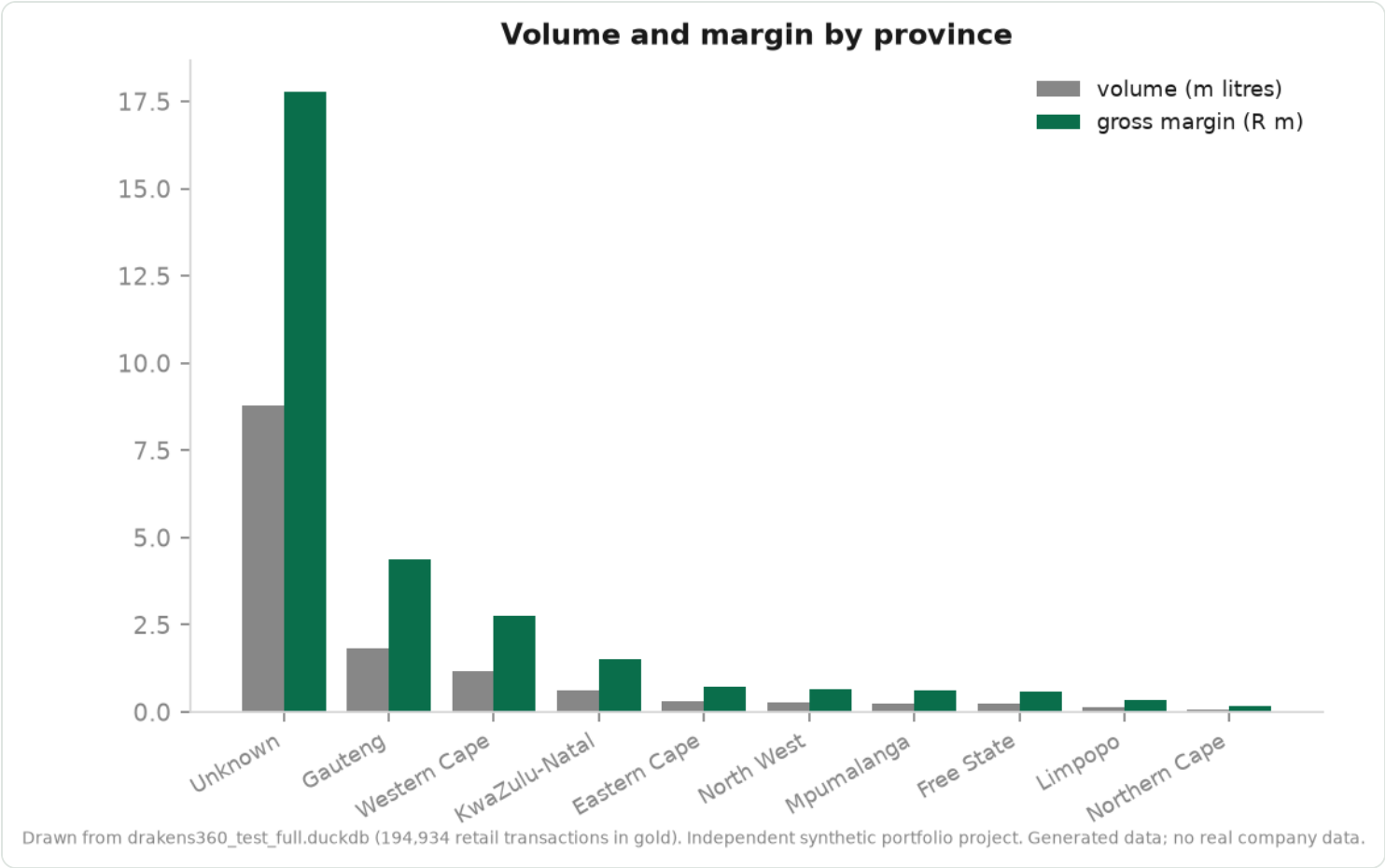


Provinces shaded by margin per site; bubble area is total margin and colour is the investment recommendation. The N1, N2 and N3 are drawn because corridor sites are protected from divestment regardless of score. Shaded areas are the convex hull of each province’s synthetic sites, not a province boundary.

| Measure                | Value          |
|------------------------|----------------|
| Budget                 | R1.50bn        |
| Funded projects        | 250            |
| Capital committed      | R1.384bn       |
| Expected annual uplift | R353.1m        |
| Portfolio ROI          | 25.5% per year |
| Blended payback        | 3.9 years      |
| Provinces covered      | 9 of 9         |

Plan by province

| Province      | Projects | Capex (R m) | Uplift (R m) |
|---------------|----------|-------------|--------------|
| Gauteng       | 84       | 460.80      | 121.48       |
| Western Cape  | 51       | 285.60      | 81.67        |
| KwaZulu-Natal | 34       | 190.80      | 42.80        |
| Eastern Cape  | 18       | 97.70       | 22.64        |
| Free State    | 16       | 87.00       | 18.99        |
| North West    | 15       | 86.40       | 20.20        |
| Mpumalanga    | 15       | 82.10       | 18.43        |
| Limpopo       | 9        | 45.20       | 15.16        |
| Northern Cape | 7        | 42.20       | 10.72        |



Volume and margin by province.

# Machine learning

Six experiments, each reported against a baseline it has to beat. Not all of them win, and all six are listed — a portfolio that shows only the models that worked is not showing the job.

| Model                  | Headline result                                                          | Verdict                                             |
|------------------------|--------------------------------------------------------------------------|-----------------------------------------------------|
| Customer Churn         | ROC-AUC 0.841, 87% precision in the top decile                           | Usable — ranks well above chance                    |
| Demand Forecast        | MAE 39.4 L against a seasonal-naive 54.4 L                               | Beats its baseline by 27.6%                         |
| Fraud Anomaly          | 1.0% of shifts flagged; cash share 29.4% against 23.9% for normal shifts | Flagged shifts differ from normal ones on both axes |
| Late Delivery          | ROC-AUC 0.564, 1.76x lift in the top decile                              | Usable — the reviewed decile is enriched            |
| Predictive Maintenance | ROC-AUC 0.616, 1.17x lift in the top decile, against a baseline of 0.620 | Does NOT beat its baseline                          |
| Stock Out Risk         | ROC-AUC 0.510                                                            | No usable signal                                    |

# What the platform found in itself

Each of these was caught by the project’s own tests, observability or baselines, which is the point of having them.

## A macro that silently corrupted every ratio

`safe_divide('a - b', 'c')` compiled to `a - (b / c)` because the macro did not parenthesise its arguments. Not an error — a plausible-looking wrong number. Caught by a range test asserting that `dim_product.base_margin_pct` falls between -5 and 100.

## A general ledger that did not balance

Journal lines were posted independently, so debits exceeded credits by 46%. Caught by `obs_reconciliation_controls`, which evaluates double-entry as data rather than only as a test. Fixed by emitting journals as balanced pairs.

### **A partitioning choice that looked obvious and was wrong**

The retail fact was partitioned on `date_key`, because every query filters on a date range. The maintenance job measured the result: 974 files averaging 0.2 MB, which OPTIMIZE can never merge because it does not cross partition boundaries. Under liquid clustering on (`date_key`, `site_id`) the same 5 million rows occupy 2 files averaging 94 MB — a 487-fold reduction in file count for identical data.

### **A leak that scored 0.998**

The stock-out model was scored against the current inventory snapshot. Its target is `days_of_cover` below a threshold; `days_of_cover` is stock over usage; and both components were features. Excluding `days_of_cover`, which the code did with a comment explaining why, excluded nothing — the model divided one component by the other and reproduced the rule. Re-pointed at the next snapshot it scores 0.510.

### **A model with no baseline, losing to one variable**

Predictive maintenance reported a 1.35x lift and passed review. It had no baseline. Ranking the queue by asset age alone scores ROC-AUC 0.620 against the model's 0.598, and the reported lift moves between 1.03 and 1.35 across capacity settings with no trend. Every experiment now computes and logs the baseline it has to beat.

### **Null foreign keys where an unknown member belonged**

The landing zone contains referential drift by design — a site code in a fact that the dimension never received. Gold left-joined those facts and kept the null key, which drops the row from every inner join and from any total grouped by province: 5,070 retail transactions with real litres and real margin were invisible to regional reporting. They now resolve to a -1 unknown member, so nothing disappears from a total and the drift is countable.

### **A sentinel that passed every non-negativity check**

The generator uses 0 as one of its numeric sentinels, and 0 satisfies every '`amount >= 0`' rule. Stripping zero the way -999 is stripped is not an option, because zero litres and zero margin are legitimate readings. The contract now checks amounts against their own row instead: a transaction of 82 litres at R22.34 reporting R0.00 of sales is internally inconsistent whatever caused it.

## **Limitations**

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Stated plainly, because a report that claims completeness is misleading.

- The Databricks results are at portfolio scale (5,000,000 retail rows through bronze, silver and gold). The DuckDB warehouse figures in this report are at test scale, and each figure is stamped with the warehouse it

came from. The two should not be read as one run.

- Predictive maintenance does not beat an age-only baseline, and stock-out risk has no signal at the current snapshot cadence. Both limits are in the generated data rather than in the estimators, and both fixes are changes to the generator: condition monitoring for the first, denser inventory readings for the second.
- The investment plan's rand figures rest on one stated benchmark — what a mid-sized site earns in fuel gross margin in a year — because the transaction fact is a sample and capital costs are not. Rank order and every ratio between sites are unaffected; only the level is rebased.
- Row filters are correct but their query cost under concurrent executive load has not been measured.
- There is no data subject access or erasure workflow. The masking and pseudonym design supports one; the process itself is not built.

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